Colouring and tiling arguments

L4-X7 Level 4 Olympiad · Combinatorics · Beyond the textbook

Kevin Hu
November 2026

Chessboard and multi-colour colourings, weights with roots of unity, invariants of tilings, constructions, and colourings of the plane.
By the end you can choose a colouring or a weight $x^a y^b$ that every tile balances, and use the weight to prove impossibility; turn the cases the weights allow into explicit tilings; force monochromatic configurations in colourings of the plane.
Prerequisites: Invariants, monovariants and extremal arguments (L4-CO1). **Contests:** BMO1, BMO2, MOG, USAJMO. **Time:** about 120 minutes.

1 Lesson

Remove two opposite corners from an 8×8 *board. Can the remaining* 62 *cells be tiled by dominoes?* Colour the board as a chessboard. Both corners are black, so 30 black and 32 white cells remain, while every domino covers one cell of each colour. So no tiling exists.

This puzzle of Max Black (1946) is the model for the whole unit. A colouring proves impossibility when every tile covers the same amount of each colour but the region does not. Read in reverse, the colouring tells you which cases to build, and the construction is usually the harder half.

1.1 Colourings and weights

Theorem 1 (weight principle).
Give each cell (a, b) a weight $w(a, b)$, a real or complex number or a residue. If a region R is tiled, the weight of R is the sum of the weights of the tiles. In particular, if every tile has weight 0, so does R.

Proof. The tiles partition R. ■

A colouring with k colours is the special case of counting colour classes. The useful weights are monomials. For a set T of cells put

$$P_T(x, y) = \sum_{(a,b)\in T} x^a y^b,$$

so the translate of T by (a, b) has polynomial $x^a y^b P_T$. Hence if $P_T(\alpha, \beta) = 0$ for every allowed orientation of every tile, then $P_R(\alpha, \beta) = 0$ for every region R that can be tiled.

Theorem 2 ($1 \times k$ strips).
An $m \times n$ board can be tiled by $1 \times k$ strips if and only if $k \mid m$ or $k \mid n$.

Proof. If $k \mid m$, put the strips down the columns. Conversely let $\zeta = e^{2\pi i/k}$ and give cell (a, b), with $0 \le a < m$ and $0 \le b < n$, the weight ζ^{a+b}. A strip starting at (a, b) has weight $\zeta^{a+b}(1 + \zeta + \cdots +$

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$\zeta^{k-1}) = 0$ in either direction, and the board has weight

$$\Big(\sum_{a=0}^{m-1}\zeta^a\Big)\Big(\sum_{b=0}^{n-1}\zeta^b\Big) = \frac{(1-\zeta^m)(1-\zeta^n)}{(1-\zeta)^2}.$$

This vanishes only if $\zeta^m = 1$ or $\zeta^n = 1$, that is $k \mid m$ or $k \mid n$. ■

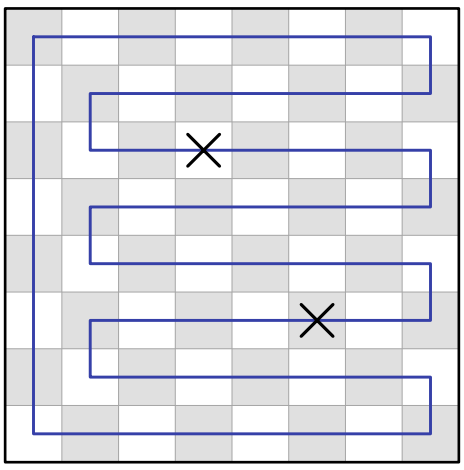
0	1	2	3	0	1
1	2	3	0	1	2
2	3	0	1	2	3
3	0	1	2	3	0
0	1	2	3	0	1
1	2	3	0	1	2

The figure shows $k = 4$ on a 6×6 board, each cell labelled by the exponent $(a + b)$ mod 4. Every strip meets each label once. The colouring by $(a + b)$ mod k also proves Theorem 2 but needs cases on m and n modulo k, which the weight replaces by one factorisation.

1.2 Constructions

When the weights allow a tiling, you still have to build one. At this level the tools are rectangles with a side divisible by k, pinwheels around a hole, and induction on the size.

Theorem 3 (Gomory).
Let $m \geq 2$ be even and $n \geq 2$. If one black and one white cell are removed from an $m \times n$ chessboard, the rest can be tiled by dominoes.

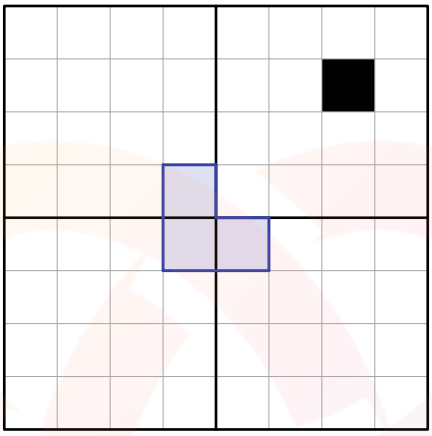


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Proof. Go down column 1 from the top-left corner, then sweep columns 2 to n row by row from the bottom up, alternating direction. Since m is even, the sweep ends next to the starting cell, so we have a closed path through every cell in which consecutive cells are neighbours. Colours alternate along the path. Removing a black and a white cell leaves at most two arcs, and between cells of opposite colours each arc has an even number of cells. Pair up consecutive cells along each arc. ■

The only idea is the closed path. Once you have the path, the rest is counting.

Theorem 4 (Golomb).
For every $n \geq 1$, a $2^n \times 2^n$ board with any one cell removed can be tiled by L-trominoes.



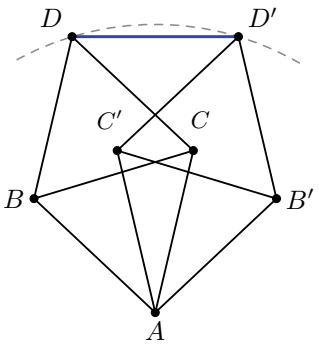
Proof. Induction on n, and $n = 1$ is one tromino. Cut the board into four $2^{n-1} \times 2^{n-1}$ quadrants. The missing cell lies in one of them, and one tromino at the centre covers a corner cell of each of the other three. Now each quadrant lacks one cell. ■

1.3 Colouring the plane

Theorem 5.
(1) If the plane is coloured with two colours, then for every $d > 0$ some two points at distance d have the same colour. (2) If the plane is coloured with three colours, some two points at distance 1 have the same colour.

Proof. (1) An equilateral triangle of side d has three vertices and there are two colours. (2) Suppose not. If ABC and BCD are unit equilateral triangles, then A, B, C have three different colours and so do B, C, D, so A and D share a colour, with $AD = \sqrt{3}$. Hence every point at distance $\sqrt{3}$ from A has the colour of A. That circle contains two points D, D' with $DD' = 1$, which is a contradiction. ■

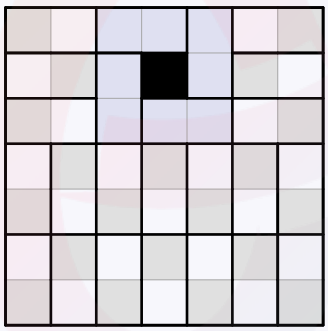
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The seven points in the figure are the *Moser spindle*. The least number of colours with no monochromatic unit pair is the chromatic number of the plane. A hexagonal pattern shows that 7 colours suffice, and de Grey proved in 2018, with a computer, that 4 do not.

1.4 Worked examples

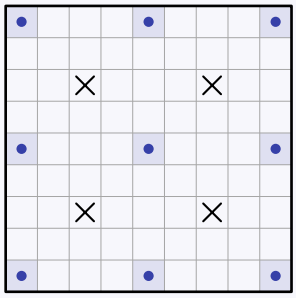
Example 1.
Let $n \geq 1$. For which cells c of a $(2n+1) \times (2n+1)$ board can the other cells be tiled by dominoes?

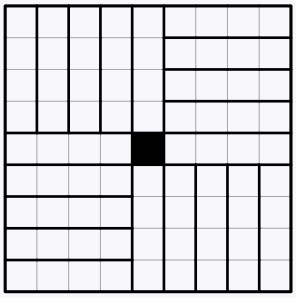


Solution.
Colour the board as a chessboard with black corners. There are $2n^2+2n+1$ black cells and $2n^2+2n$ white ones, so c must be black. Conversely let $c = (a, b)$, rows and columns numbered from 0, with $a+b$ even.
If a and b are even, the a rows above c and the $2n-a$ rows below take vertical dominoes, and the cells of row a to the left and to the right of c come in even numbers. If a and b are odd, the 3×3 block centred at c lies inside the board and its outer ring takes four dominoes. Above and below this block there are $a-1$ and $2n-1-a$ rows, both even, and beside the block the three rows have $b-1$ and $2n-1-b$ cells, both even. So every black cell works.
Remark. The colouring gives the answer in one line. The real work is the construction, split by the parity of a.

Example 2.
For which cells of a 9×9 board can the other 80 cells be tiled by 1×4 strips?

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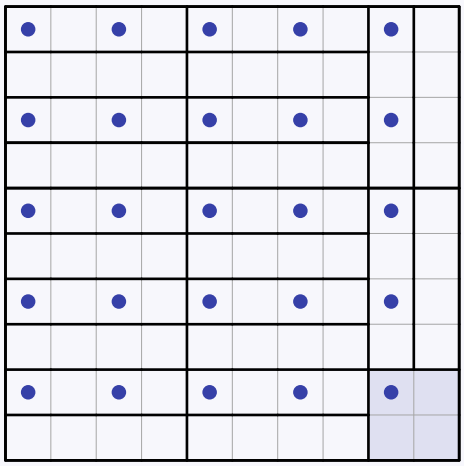


Solution.
Number rows and columns from 0 to 8 and let the missing cell be (a,b). Start with the obvious colouring of (r,s) by $(r+s)$ mod 4. The classes have $21,20,20,20$ cells and each strip covers one cell of each class, so $a+b \equiv 0 \pmod 4$. The colouring by $(r-s)$ mod 4 gives $a-b \equiv 0 \pmod 4$. That leaves the nine cells with $a,b \in \{0,4,8\}$ and also the four cells with $a,b \in \{2,6\}$, crossed in the figure. Two colourings are not enough, and the next step is the hard one.
Now use the weight $x^r y^s$ with $x,y \in \{i,-1,-i\}$. Both orientations of a strip have weight 0, as $1+t+t^2+t^3=0$ for these t. Since $t^4=1$ and $t \neq 1$,

$$1+t+\cdots+t^8 = \frac{1-t^9}{1-t} = 1,$$

so the region has weight $1 \cdot 1 - x^a y^b$, and a tiling forces $x^a y^b = 1$. With $x=i$ and $y=-1$ this says $i^a = (-1)^b$, so a is even. Similarly b is even, and then $i^a = (-1)^b = 1$ gives $4 \mid a$, and by symmetry $4 \mid b$.
All nine cells work. For $(0,0)$ or $(0,4)$, row 0 minus the cell takes two strips and the 8×9 block below takes vertical strips. The centre is the pinwheel of four 4×5 blocks in the figure, and the other cells follow by symmetry.
Remark. The weight $i^r(-1)^s$ mixes a 4-colouring of the rows with a 2-colouring of the columns. No colouring by $r+s$ or $r-s$ can see that.

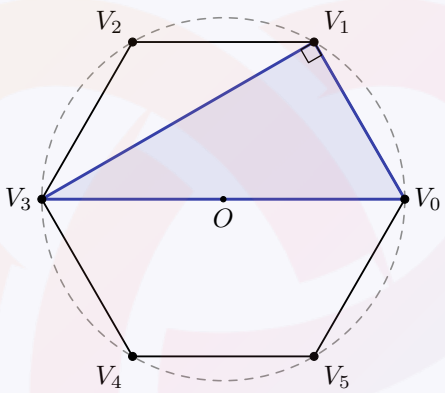
Example 3.
A 10×10 board is tiled by 1×4 strips and 2×2 squares. Prove that the number of squares is odd, and find the least possible number of squares.



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Solution.
Mark the 25 cells whose row and column numbers, counted from 0, are both even. A 2×2 square covers exactly one marked cell. A horizontal strip covers four consecutive columns, two of them even, so the strip covers two marked cells if its row is even and none otherwise. The same holds for vertical strips. So the number of squares has the parity of 25, and is odd. One square is enough. Tile columns 0 to 7 by horizontal strips, rows 0 to 7 of columns 8 and 9 by vertical strips, and put the square in the corner. So the least number is 1.
Remark. Theorem 2 forbids a tiling by strips alone, and the marked cells say exactly why. For the same reason, no region has two tilings by these pieces in which the numbers of squares differ by 1.

Example 4.
The plane is coloured red and blue. Prove that some triangle with angles 30°, 60°, 90° and hypotenuse 2 has its three vertices of the same colour.



Solution.
By Theorem 5(1) there are points V_0, V_3 of the same colour, say red, at distance 2. Let $V_0V_1 \dots V_5$ be the regular hexagon with diameter V_0V_3. For $k \neq 0, 3$ the triangle $V_0V_kV_3$ has a right angle at V_k and one leg equal to the side 1, so the triangle has the required angles. If any V_k with $k \neq 0, 3$ is red, we are done. Otherwise V_1, V_2, V_4, V_5 are blue, and $V_1V_2V_4$ is such a triangle with diameter V_1V_4. ■
Remark. The triangle cannot be equilateral. Colour the half-open strips $\frac{\sqrt{3}}{2}k \leq y < \frac{\sqrt{3}}{2}(k+1)$ alternately. For a unit equilateral triangle the middle vertex is within $\frac{\sqrt{3}}{2}$ vertically of each of the others, while the top and bottom vertices are at least $\frac{\sqrt{3}}{2}$ apart, so the three vertices never share a colour.

Common traps.

- A balanced colouring proves nothing about existence. Example 2 has cells that pass two colourings and still fail.
- State the colouring or weight exactly, check every orientation of every tile, and say which roots of unity you use.
- In the plane the colour classes are arbitrary sets, so the argument must use finitely many points.

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1.5 How to spot a colouring argument

1. A region and a set of tiles, with the question whether the region can be tiled, or which cells can be removed.
2. A tile or a move that always covers a fixed pattern of cells. Compute what one tile does to each colour class.
3. A question about the parity or the least number of tiles of one kind.
4. Two colourings that disagree, or one that is not strong enough. Try $x^a y^b$ with x and y roots of unity of different orders.
5. A colouring of the whole plane. Look for a finite configuration (a triangle, a hexagon, a rhombus) that forces two equal colours.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ (a) A rectangle is tiled by tetrominoes, none of which is a T-tetromino. Prove that the number of L-tetrominoes (counting both mirror images) is even.

(b) Tile a 3×4 rectangle by one L-tetromino and two T-tetrominoes.

Problem 2 ⋆⋆⋆ A box of size $2 \times 2 \times n$, where $n \geq 1$, is filled with $2n$ bricks of size $1 \times 1 \times 2$. A *fault plane* is a plane parallel to a face of the box that cuts the box into two parts without cutting any brick.

(a) Prove that there is always a fault plane.

(b) Show that a $2 \times 3 \times 3$ box can be filled with $1 \times 1 \times 2$ bricks so that there is no fault plane.

Problem 3 ⋆⋆⋆ A $2n \times 2n$ board is tiled by dominoes. Prove that exactly n of the dominoes that cover a cell of the main diagonal (from the top-left corner to the bottom-right corner) have their other cell above the diagonal.

Problem 4 ⋆⋆⋆⋆ Let $n > 1$ be an integer coprime to 6. One cell is removed from an $n \times n$ board. Determine all cells for which the remaining cells can be tiled by 2×3 rectangles (in either orientation).

Problem 5 ⋆⋆⋆⋆ A knight visits every cell of a $4 \times n$ board exactly once, by knight's moves, and does not return to its first cell. Prove that the first and last cells both lie in the top row or the bottom row, and that exactly one move of the tour joins two cells of the middle rows.

Problem 6 ⋆⋆⋆⋆ Let $n \geq 3$ be odd. For which n can the $n \times n \times n$ cube with its central unit cube removed be filled with $1 \times 1 \times 4$ bricks?

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

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1. **AMC 8 2024 Problem 7** ⋆⋆
   Colour the columns alternately and compare what each large tile covers.

2. **JBMO 2006 Problem 4** ⋆⋆
   The chessboard colouring gives an upper bound. Then find a placement that reaches the bound.

3. **IMO 1983 Problem 4** ⋆⋆⋆
   Look at the points that divide each side into thirds.

4. **Baltic Way 2007 Problem 7** ⋆⋆⋆
   Area gives one condition. For the rest, colour the unit triangles like a chessboard and count.

5. **IMO Shortlist 2002 C2** ⋆⋆⋆
   Find many black squares no two of which can be covered by the same tromino.

6. **EGMO 2019 Problem 2** ⋆⋆⋆
   Count pairs of a cell and an adjacent covered cell in two ways, and look for a repeating pattern for the construction.

7. **Balkan MO 2000 Problem 3** ⋆⋆⋆
   For the bound, weight or colour the rectangle so that each piece meets a fixed amount of the weight.

8. **IMO 1999 Problem 3** ⋆⋆⋆⋆
   Split the board into nested frames, and colour so that the condition separates into two independent halves.

9. **IMO 1993 Problem 3** ⋆⋆⋆⋆
   Colour the cells with three colours along diagonals and follow the parities of the three counts.

10. **IMO 2004 Problem 3** ⋆⋆⋆⋆
    First show that the hooks pair up into larger rectangles, then use colourings to restrict the sides.

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4 Hints

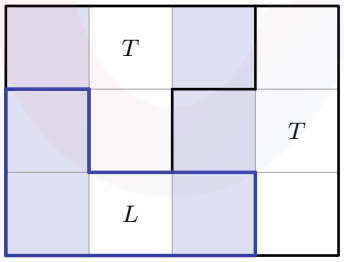
1. Colour the columns alternately, along the side of even length. Compute what each kind of tetromino does to the difference of the two colours, modulo 4.
2. How many of the planes inside the box does one brick cross? Show that a plane which is not a fault plane is crossed by at least two bricks.
3. Colour the board so that the diagonal is black. Count black and white cells above the diagonal, not on the diagonal.
4. Weights $x^r y^s$ with (x, y) equal to $(-1, -1)$, (ω, ω) and (ω, ω^2) give three congruences. They do not rule out every bad cell, so look directly at a removed cell on the edge.
5. Where can a knight go from a cell of the top or bottom row? Combine the answer with the chessboard colouring.
6. Use the weight $x^a y^b z^c$ with $x, y, z \in \{i, -1, -i\}$, and treat $n \equiv 1$ and $n \equiv 3 \pmod 4$ separately. For the construction, cut the cube into two slabs and a middle layer.

5 Solutions

Problem 1

(a) A rectangle is tiled by tetrominoes, none of which is a T-tetromino. Prove that the number of L-tetrominoes (counting both mirror images) is even.

(b) Tile a 3×4 rectangle by one L-tetromino and two T-tetrominoes.



(a) The chessboard colouring sees nothing, since an L-tetromino covers two cells of each colour, so we colour by columns. The area is a multiple of 4, so one side is even. Turn the rectangle so that the number of columns is even, and colour the columns alternately black and white. Then the rectangle has as many black cells as white cells.

For a tile let d be the number of its black cells minus the number of its white cells. A 2×2 square has $d = 0$, and a 1×4 strip has $d = 0$ lying across the columns and $d = \pm 4$ lying in one column. An S-tetromino meets two adjacent columns in 2 cells each, or three columns in $1, 2, 1$ cells, so $d = 0$. An L-tetromino meets two adjacent columns in 3 and 1 cells, or three columns in $2, 1, 1$ or $1, 1, 2$ cells, where the end columns have the same colour, so $d = \pm 2$.

The values of d add up to 0. Modulo 4 every term is 0 except the ℓ terms ± 2 from the L-tetrominoes, so $2\ell \equiv 0 \pmod 4$. Hence ℓ is even. ■

(b) The figure shows such a tiling.

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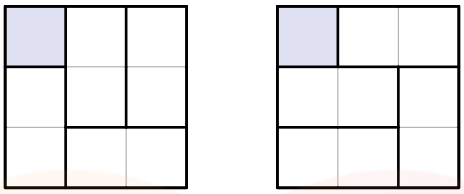
Remark. A T-tetromino has $d = \pm 2$ in one column and $d = 0$ across three, which is what (b) exploits. The chessboard colouring, under which only the T-tetromino is unbalanced, shows that the number of T-tetrominoes is always even.

Problem 2

A box of size $2 \times 2 \times n$, where $n \geq 1$, is filled with $2n$ bricks of size $1 \times 1 \times 2$. A fault plane *is a plane parallel to a face of the box that cuts the box into two parts without cutting any brick.*

(a) Prove that there is always a fault plane.

(b) Show that a $2 \times 3 \times 3$ box can be filled with $1 \times 1 \times 2$ bricks so that there is no fault plane.



(a) The planes in question are the plane halving each of the two edges of length 2 and the $n - 1$ planes across the edge of length n, so $n + 1$ planes in all. The two cells of a brick are neighbours in exactly one direction, so each brick crosses exactly one of these planes.

Take one of the planes. Both parts have even volume, namely $2n$ for a halving plane and $4j$ for a cross plane. A brick that does not cross the plane covers two cells of one part, and a brick that crosses the plane covers exactly one cell of each part. So the number of bricks crossing the plane is even. If no plane is a fault plane, every plane is crossed by at least 2 bricks, and counting crossings gives

$$2n = \#\{\text{bricks}\} \geq 2(n + 1),$$

which is false. So some plane is a fault plane.

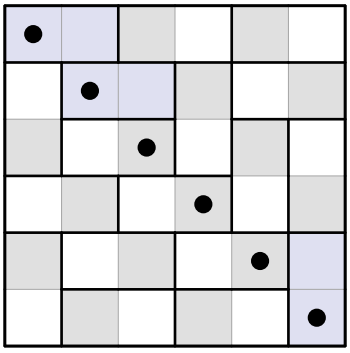
(b) This part is trial and error. Use the two 3×3 layers in the figure, each with four flat bricks, and one vertical brick through the shaded corner cells. The plane between the layers is crossed by the vertical brick. In the lower layer the two vertical dominoes in the top rows cross the line between the first two rows, the domino in the first column crosses the line between rows 2 and 3, and the domino in the bottom row crosses the line between columns 2 and 3. In the upper layer the domino in the middle row crosses the line between columns 1 and 2. So none of the five planes is a fault plane. ■

Remark. For a $2 \times 3 \times n$ box there are $n + 2$ planes against $3n$ bricks, so the count gives nothing, and (b) shows that no argument could succeed there.

Problem 3

A $2n \times 2n$ board is tiled by dominoes. Prove that exactly n of the dominoes that cover a cell of the main diagonal (from the top-left corner to the bottom-right corner) have their other cell above the diagonal.

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Number rows and columns from 1 to $2n$ and colour (r,s) black when $r+s$ is even, so the diagonal cells (r,r) are black. No two diagonal cells are neighbours, so the $2n$ diagonal cells lie in $2n$ different dominoes, and the other cell of such a domino is $(r,r\pm1)$ or $(r\pm1,r)$, which is white.

The hard step is to see that only the dominoes meeting the region above the diagonal matter. Let U be the set of cells (r,s) with $s>r$. A neighbour of a cell of U is in U or on the diagonal. So every domino meeting U either lies inside U, and covers one cell of each colour, or is a diagonal domino whose other cell is a white cell of U. Hence the number of diagonal dominoes pointing into U is $W-B$, where W and B count the white and black cells of U.

The cells of U with $s-r=d$ number $2n-d$, and they are black exactly when d is even. So

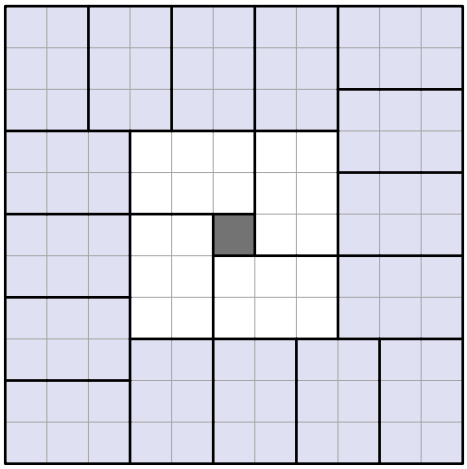
$$W-B=\sum_{d=1}^{2n-1}(-1)^{d+1}(2n-d)=\big((2n-1)-(2n-2)\big)+\cdots+(3-2)+1=n,$$

since there are $n-1$ brackets, each equal to 1. ■

Remark. The colouring turns a question about one tiling into a count over the region, which is why the answer does not depend on the tiling.

Problem 4

Let $n>1$ be an integer coprime to 6. One cell is removed from an $n\times n$ board. Determine all cells for which the remaining cells can be tiled by 2×3 rectangles (in either orientation).



Number rows and columns from 0 to $n-1$, let (a,b) be the removed cell and $c=\frac{n-1}{2}$. We show that a tiling exists if and only if (a,b) is not on the edge, $a\equiv b\equiv c\pmod 3$ and $a\equiv b\pmod 2$.

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There are three parts. The edge argument is the one most solvers miss, because the weights look as if they should be enough. The construction is the longest part, but each step is routine.

Edge cells fail. By symmetry let the removed cell be $(0, b)$. The tile covering $(1, b)$ would contain $(0, b)$ if the tile reached row 0, so the tile starts in row 1. Every tile has both sides at least 2, so this tile also covers $(1, b-1)$ or $(1, b+1)$, say $(1, b+1)$. The tile covering $(0, b+1)$ meets row 0, so in the same way the tile starts in column $b+1$, and the tile covers $(1, b+1)$ too. This is a contradiction.

Weights. Give cell (r, s) the weight $x^r y^s$ and put $S(t) = 1 + t + \cdots + t^{n-1}$. A tile has weight $x^r y^s(1+x)(1+y+y^2)$ or $x^r y^s(1+x+x^2)(1+y)$, which vanishes for $x = y = -1$ and for $x, y \in \{\omega, \omega^2\}$, where $\omega = e^{2\pi i/3}$. By Theorem 1 a tiling forces

$$S(x)S(y) = x^a y^b$$

for these (x, y). As n is odd, $S(-1) = 1$, so $a \equiv b \pmod 2$. As $3 \nmid n$, ω^n is ω or ω^2, and $|S(\omega)| = \frac{|1-\omega^n|}{|1-\omega|} = 1$. Also $S(\omega) = \omega^c D$ with $D = \sum_{j=-c}^{c} \omega^j$ real, so $D = \pm 1$. Now $(x, y) = (\omega, \omega^2)$ gives $\omega^{a-b} = S(\omega)S(\omega^2) = |S(\omega)|^2 = 1$, and $(x, y) = (\omega, \omega)$ gives $\omega^{a+b} = S(\omega)^2 = \omega^{2c} D^2 = \omega^{2c}$. Hence $a \equiv b$ and $2a \equiv 2c \pmod 3$, so $a \equiv b \equiv c \pmod 3$.

Construction. A $p \times q$ rectangle with $6 \mid p$ and $q \geq 2$ splits into 6×2 and 6×3 blocks, as $q = 2s + 3t$ with $s, t \geq 0$, so the rectangle can be tiled. A $3 \times q$ rectangle with q even can be tiled too. Let Q_d be the square of side $2d+1$ without its centre. Then Q_2 is four tiles in a pinwheel, and Q_3 is four 3×4 rectangles in a pinwheel. A pinwheel of four $3 \times (2d+4)$ rectangles around Q_d gives Q_{d+3}, as in the figure. So Q_d can be tiled whenever $d \geq 2$ and $d \not\equiv 1 \pmod 3$.

Now let (a, b) satisfy the conditions. Then $a \equiv b \pmod 6$, and $c \equiv a \pmod 3$ gives $n - 1 - a = 2c - a \equiv a \pmod 6$, and the same for b. Choose $d \in \{2, 3, 5, 6\}$ with $d \equiv a \pmod 6$, which is possible as c is 0 or 2 modulo 3. The numbers a, b, $n-1-a$, $n-1-b$ are positive and congruent to d modulo 6, so all are at least d. Hence the square of side $2d+1$ centred at (a, b) fits in the board, and a copy of Q_d fills the square apart from (a, b). The rows above and below the square number $a - d$ and $n - 1 - a - d$, both divisible by 6, and the blocks to the left and right have $2d + 1 \geq 5$ rows and $b - d$ and $n - 1 - b - d$ columns, both divisible by 6. Each of these rectangles can be tiled. ■

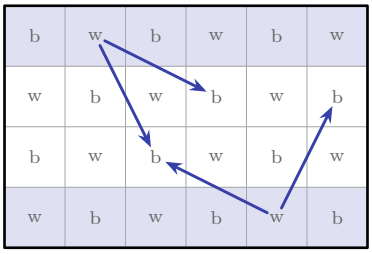
Remark. For $n = 7$ the four corners pass all three weight tests, and only the edge argument removes them. This is the first trap in the lesson.

Answer: With rows and columns numbered from 0 to $n-1$ and $c = \frac{n-1}{2}$, the cells (a, b) not on the edge of the board with $a \equiv b \equiv c \pmod 3$ and $a \equiv b \pmod 2$

Problem 5

A knight visits every cell of a $4 \times n$ board exactly once, by knight's moves, and does not return to its first cell. Prove that the first and last cells both lie in the top row or the bottom row, and that exactly one move of the tour joins two cells of the middle rows.

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Call the cells of the top and bottom rows *outer* and the others *inner*. There are $2n$ of each. A knight's move changes the row by 1 or 2, so from row 1 the knight reaches row 2 or 3, and from row 4 the knight reaches row 3 or 2. Hence two consecutive cells of the tour are never both outer. Also a knight's move always changes the colour of the cell. The first claim is the hard one, and the second is a short count once we have the first.

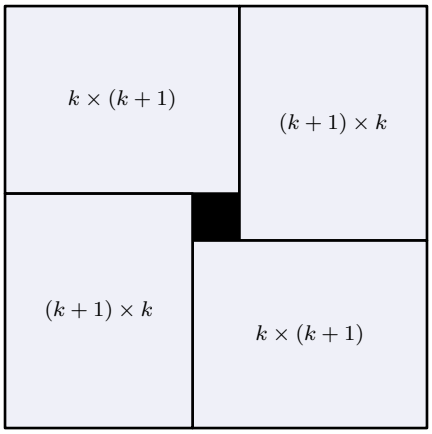
Write the tour as $c_1, c_2, \ldots, c_{4n}$. Suppose c_1 is inner. Then the $2n$ outer cells occupy positions among $2, 3, \ldots, 4n$, no two consecutive. A set of pairwise non-consecutive numbers from these $4n-1$ numbers has at most $2n$ elements, with equality only for $\{2, 4, \ldots, 4n\}$. So the outer cells are $c_2, c_4, \ldots, c_{4n}$, all of the same colour as c_2. But the top row contains cells of both colours, since $n \geq 2$ (a knight cannot move on a 4×1 board). This is a contradiction, so c_1 is outer, and in the same way c_{4n} is outer.

Now the outer cells include c_1 and c_{4n}, and between consecutive outer cells of the tour there is at least one inner cell. That makes $2n-1$ gaps holding $2n$ inner cells in total, so exactly one gap holds two inner cells and the others hold one. A move joins two inner cells only inside that gap, so there is exactly one such move. ■

Remark. For a closed tour the same argument forces the tour to alternate outer and inner cells, hence all outer cells to share a colour. This is Pósa's proof that a $4 \times n$ board has no closed knight's tour.

Problem 6

Let $n \geq 3$ be odd. For which n can the $n \times n \times n$ cube with its central unit cube removed be filled with $1 \times 1 \times 4$ bricks?



Use coordinates $0 \leq a, b, c \leq n-1$ and write $n = 2k+1$, so the hole is (k, k, k). Give the unit cube (a, b, c) the weight $x^a y^b z^c$ with $x, y, z \in \{i, -1, -i\}$. A brick has weight 0, because for each of these t we have $1 + t + t^2 + t^3 = 0$. Let $S(t) = 1 + t + \cdots + t^{n-1} = \frac{1-t^n}{1-t}$. By Theorem 1, a filling forces

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$$S(x)S(y)S(z) = (xyz)^k \quad \text{for all } x, y, z \in \{i, -1, -i\}.$$

Case $n \equiv 3 \pmod 4$. Then $t^n = t^3$ and $S(t) = 1 + t + t^2$, so $S(i) = i$ and $S(-1) = 1$. With $x = y = i$ and $z = -1$ the left side is $i \cdot i \cdot 1 = -1$, while $xyz = 1$ makes the right side 1. So no filling exists.

Case $n \equiv 1 \pmod 4$. This is the case where the third dimension matters. Then $t^n = t$ and $S(t) = 1$, so we need $(xyz)^k = 1$. With $x = i$ and $y = z = -1$ this says $i^k = 1$, so $4 \mid k$ and $n = 2k + 1 \equiv 1 \pmod 8$. Hence $n \equiv 5 \pmod 8$ is impossible too.

Construction for $n \equiv 1 \pmod 8$. Now $4 \mid k$. The layers $c < k$ form an $n \times n \times k$ box, filled by bricks along the third axis, and so do the layers $c > k$. The middle layer $c = k$ is an $n \times n$ square without its centre. Cut that square into four rectangles of size $k \times (k + 1)$ as in the figure, namely rows 0 to $k - 1$ with columns 0 to k, rows 0 to k with columns $k + 1$ to $2k$, and the two images of these under the half-turn about the centre. Each rectangle has a side k divisible by 4, so strips fill the rectangle. ■

Remark. In the plane, the square without its centre can be tiled for $n \equiv 7 \pmod 8$ as well, by the same pinwheel with $4 \mid k + 1$. The weight with $x = y = i$ and $z = -1$ has no planar analogue, and that is what rules out $n \equiv 7$ here.

Answer: $n \equiv 1 \pmod 8$

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